

MOURGASH, BulgariaWMO: **156000**Lat: **42.833N**Lon: **23.667E**Elev: **5551**StdP: **11.98**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.6	6.4	-5.2	5.1	8.6	-0.3	6.6	11.1	76.8	30.0	75.5	31.3	29.3	0	0.956

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.7	71.8	58.5	68.9	57.6	66.4	56.8	61.6	67.1	60.0	65.6	58.5	63.9	13.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.9	94.8	63.3	58.0	88.6	62.0	56.4	83.6	60.7	30.4	67.4	29.2	65.9	28.0	64.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
63.5	54.3	45.1														
			DB	-1.8	78.0	4.9	5.0	-5.4	81.6	-8.2	84.6	-11.0	87.4	-14.5	91.1	
			WB	-2.3	65.6	4.8	2.1	-5.8	67.1	-8.6	68.3	-11.3	69.5	-14.8	71.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	23.8	24.6	29.5	37.9	46.6	52.8	56.8	57.4	49.9	42.1	35.3	26.6
	DBStd	14.40	9.47	9.41	8.75	7.91	6.74	6.60	5.93	5.93	7.41	8.76	9.11	9.17
	HDD50	4266	812	712	636	369	146	45	10	8	90	267	446	725
	HDD65	9008	1277	1132	1101	812	569	367	262	243	454	709	892	1190
	CDD50	755	0	0	0	8	42	130	221	239	87	24	4	0
	CDD65	20	0	0	0	0	0	2	8	9	1	0	0	0
	CDH74	35	0	0	0	0	1	1	20	11	2	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
	WSAvg	19.1	22.7	23.3	22.1	19.6	17.0	15.3	15.2	14.7	16.6	18.1	22.3	23.0
	PrecAvg	29.0	1.6	1.5	1.8	2.2	3.9	3.6	3.1	2.4	2.5	2.2	1.4	1.6
Precipitation	PrecMax	47.6	3.7	2.5	4.0	4.6	8.1	7.4	7.8	11.1	8.1	4.6	3.3	3.6
	PrecMin	20.4	0.4	0.5	0.5	0.4	1.2	1.4	0.6	0.3	0.4	0.2	0.2	0.3
	PrecStd	6.6	0.8	0.7	0.8	1.1	1.8	1.7	1.9	2.2	1.9	1.2	0.7	0.7
	PrecStd	6.6	0.8	0.7	0.8	1.1	1.8	1.7	1.9	2.2	1.9	1.2	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	45.1	48.6	52.6	61.5	66.2	72.1	77.2	75.7	71.1	63.8	58.9	49.2	42.1
	0.4% MCWB	36.3	39.3	41.8	48.6	55.5	60.3	57.4	59.4	56.2	51.1	45.8	37.2	32.0
	2% DB	41.3	43.2	48.0	56.5	61.5	67.9	71.9	71.7	66.4	59.2	52.7	44.4	37.2
	2% MCWB	34.2	35.4	40.5	46.1	52.8	58.8	57.6	59.2	54.7	49.4	42.8	36.0	32.0
	5% DB	38.3	39.3	44.3	52.2	58.7	64.8	68.6	68.7	62.8	55.9	49.2	40.9	36.8
	5% MCWB	33.8	34.5	38.0	44.3	51.7	57.6	57.6	57.4	53.1	48.3	41.7	35.9	32.0
	10% DB	35.6	35.9	41.3	48.6	56.1	62.3	65.7	66.2	59.7	53.0	46.1	38.0	32.0
	10% MCWB	32.5	33.1	36.5	42.7	50.1	56.2	57.2	56.6	51.8	46.5	40.3	34.1	32.0
	0.4% WB	39.1	40.5	44.4	50.4	57.7	63.5	63.5	63.7	58.3	55.3	48.7	42.1	37.2
	0.4% MCDB	42.6	46.7	50.5	56.3	62.2	67.2	68.1	68.9	66.1	60.5	55.0	43.1	37.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	36.3	37.3	41.2	47.6	54.7	60.9	61.2	61.2	56.1	51.7	45.8	39.3	32.0
	2% MCDB	38.9	41.4	46.2	53.7	59.5	65.7	67.0	67.1	63.1	56.2	49.7	41.6	32.0
	5% WB	34.5	35.2	39.0	45.5	52.8	58.7	59.8	59.6	54.4	49.8	43.9	36.8	32.0
	5% MCDB	36.9	37.7	42.8	50.8	57.3	63.2	65.4	65.6	60.6	53.7	47.1	39.4	32.0
	10% WB	32.8	33.4	37.1	43.4	51.0	56.9	58.4	58.1	52.9	48.0	41.8	34.9	32.0
	10% MCDB	35.1	35.7	40.8	48.2	55.0	61.5	63.5	64.0	58.2	51.7	44.5	37.3	32.0
Mean Daily Temperature Range	MDBR	7.9	8.2	8.9	9.4	9.4	9.4	10.1	9.7	9.2	9.1	7.9	7.7	7.7
	5% DB MCDBR	9.8	11.0	12.0	13.5	12.9	12.0	14.0	12.4	13.1	12.5	10.4	9.2	9.2
	5% DB MCWBR	7.3	7.4	7.6	8.4	9.2	9.2	8.7	7.7	7.2	7.8	6.7	6.6	6.6
	5% WB MCDBR	8.5	9.0	10.9	11.9	12.0	10.8	11.4	10.8	11.4	11.4	9.1	7.7	7.7
	5% WB MCWBR	7.0	7.0	7.8	8.5	9.5	9.4	8.7	8.1	7.5	7.9	6.7	6.6	6.6
	5% WB MCWBR	7.0	7.0	7.8	8.5	9.5	9.4	8.7	8.1	7.5	7.9	6.7	6.6	6.6
Clear-Sky Solar Irradiance	taub	0.260	0.280	0.312	0.352	0.345	0.345	0.347	0.354	0.335	0.306	0.280	0.262	0.262
	taud	2.468	2.445	2.428	2.324	2.378	2.393	2.378	2.360	2.413	2.500	2.546	2.507	2.507
	Ebn at Noon	285	296	297	290	294	293	291	286	284	281	275	274	274
	Edh at Noon	23	28	32	38	37	37	37	37	32	26	21	20	20
All-Sky Solar Radiation	RadAvg	501	758	1088	1432	1717	1931	1995	1801	1324	897	561	420	420
	RadStd	75	98	115	167	144	160	148	153	106	112	62	59	59